

MyHomeBase™

Contractor Module

Developer Reference

Architecture · API Reference · Database Schema · Annotated Code

GENERATED	June 26, 2026
STACK	React 18 + TypeScript (Vite) / Express 5 + Drizzle ORM (PostgreSQL)
MONOREPO	pnpm workspaces — @workspace/myhomebase, @workspace/api-server, @workspace/db
PRODUCTION	gothomebase.com

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MyHomeBase™ — Contractor Module

Developer Reference: Architecture, API, Schema & Annotated Code

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Routes version: 2025-11-02-21:28

Source sizes: routes.ts 18,083 lines, schema.ts 2,377 lines

Stack: React 18 + TypeScript (Vite), Express 5 + Drizzle ORM (PostgreSQL)

Monorepo: pnpm workspaces — @workspace/myhomebase (frontend), @workspace/api-server (backend), @workspace/db (schema)

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1. Module Overview

The contractor module is a B2B vertical within MyHomeBase™ serving three role types:

Role	companyRole value	Capabilities
Owner	'owner'	Full profile, billing, team management, all CRM
Admin	'admin'	Company-level admin; most owner rights except billing
Tech	'tech'	Invoice upload, own job history only

Routes extracted from live source: 111 contractor endpoints total.

The module covers:

- Profile & Discovery — public-facing listing, reviews, ratings
- Lead Capture — homeowner messaging, proposals, connection codes
- Team Management — invite/suspend/remove techs; seat limits enforced server-side
- CRM (Pro only) — clients, jobs, quotes, invoices, Stripe payment links
- Stripe Connect — direct payout onboarding; charges-enabled check before invoicing
- Boost — paid geo-radius visibility boosts (Stripe charge at purchase)
- Personal Home — contractors may track one personal property

2. Subscription Tiers

Source: artifacts/api-server/src/routes/routes.ts — CONTRACTOR_PLANS constant (auto-seeded to subscription_plans table at startup)


```

TYPESCRIPT
const CONTRACTOR_PLANS = [
  {
    tierName: 'contractor_basic',
    displayName: 'Contractor Basic',
    description: 'Essential tools for independent contractors',
    monthlyPrice: '20.00',
    minHouses: 0,
    maxHouses: 1, // 1 personal home for maintenance tracking
    planType: 'contractor',
    features: ['Get found by homeowners', 'Messaging with homeowners', 'Send proposals', 'Reviews and ratings'],
    profile: 'Far up to $20/month in referral credits',
    referralCreditCap: 20.00,
    hasCrmAccess: false,
    sortOrder: 0
  },

```

Grandfathered Emails (from live source)

```

TYPESCRIPT
const GRANDFATHERED_EMAILS = (process.env.GRANDFATHERED_EMAILS ||
'lihandyman2008@gmail.com,bryanmendezdesign@gmail.com,freshandcleangutters@gmail.com').split(',').map(e =>
e.trim().toLowerCase()).filter(Boolean);

```

Trial / Subscription State Machine

The users.subscriptionStatus field drives access:

```

trialing  !' active      (Stripe webhook: checkout.session.completed)
trialing  !' past_due   (trial ended, no payment)
active    !' past_due   (invoice.payment_failed)
past_due  !' canceled   (after grace period)
any       !' grandfathered (manual admin override; bypasses all checks)

```

3. Authentication & Authorization

Session Shape

Express-session populated on login (Replit OIDC !' replitAuth.ts, Google OAuth !' googleAuth.ts, email/password !' storage.ts).

```

TYPESCRIPT
// req.session.user
{
  id: string;           // users.id
  email: string;
  role: 'contractor' | 'homeowner' | 'agent';
  companyId?: string;
  companyRole?: 'owner' | 'admin' | 'tech';
  firstName?: string;
  lastName?: string;
  subscriptionStatus?: string;
  trialEndsAt?: Date;
}

```

Middleware Chain

```

isAuthenticated          - session check; 401 if missing
requireRole('contractor') - role check; 403 if mismatch
requireContractorSubscription - trial/active check
requireCompanyRole(roles) - checks req.session.user.companyRole
requireNotSuspended()    - checks suspendedUserIds Set
requireSameCompany        - cross-company data isolation

```

`requireContractorSubscription` (from live source)


```

TYPESCRIPT
const requireContractorSubscription = async (req: any, res: any, next: any) => {
  try {
    if (!req.session?.isAuthenticated || !req.session?.user) {
      return res.status(401).json({ message: 'Unauthorized' });
    }

    const userId = req.session.user.id;
    const user = await storage.getUser(userId);

    if (!user) {
      return res.status(401).json({ message: 'User not found' });
    }

    // Skip check for non-contractors
    if (user.role !== 'contractor') {
      return next();
    }

    // Check if grandfathered - free unlimited access
    const isGrandfathered = user.email && GRANDFATHERED_EMAILS.includes(user.email.toLowerCase());
    if (isGrandfathered || user.subscriptionStatus === 'grandfathered') {
      return next();
    }

    // Check if active subscription
    if (user.subscriptionStatus === 'active') {
      return next();
    }

    // Check if still in trial
    if (user.subscriptionStatus === 'trialing' && user.trialEndsAt) {
      const trialEnd = new Date(user.trialEndsAt);
      if (trialEnd > new Date()) {
        return next(); // Still in trial
      }
    }

    // Trial expired or no subscription - contractors have NO free features
    return res.status(403).json({
      message: 'Subscription required',
      code: 'SUBSCRIPTION_REQUIRED',
      detail: 'Your free trial has ended. Contractors must subscribe to access HomeBase features.'
    });
  } catch (error) {
    console.error('[CONTRACTOR SUBSCRIPTION CHECK] Error:', error);
    return res.status(500).json({ message: 'Failed to verify subscription' });
  }
};

```

`requireAdmin` (from live source)

Reads ADMIN_EMAILS env var (comma-separated). Protects /api/admin/* routes.

```

TYPESCRIPT
const requireAdmin: any = (req: any, res: any, next: any) => {
  if (!req.session?.isAuthenticated || !req.session?.user) {
    return res.status(401).json({ message: "Unauthorized" });
  }

  const adminEmails = (process.env.ADMIN_EMAILS || '').split(',').map(e => e.trim()).filter(Boolean);
  if (!adminEmails.includes(req.session.user.email)) {
    return res.status(403).json({ message: "Forbidden - admin access required" });
  }

  next();
};

```


4. File Map

Backend (`artifacts/api-server/src/`)

```

routes/routes.ts      - All Express route registrations (18,083 lines)
replitAuth.ts         - Replit OIDC setup + isAuthenticated, requireRole, etc.
googleAuth.ts         - Google OAuth setup
storage.ts            - IStorage interface + DatabaseStorage implementation
objectStorage.ts      - Replit Object Storage wrapper
invoice-analysis-service.ts - OpenAI-powered PDF invoice extraction
geocoding-service.ts  - Nominatim + haversine distance
security-audit.ts     - auditLogger, sessionManager
email-service.ts      - SendGrid wrapper
sms-service.ts        - Twilio wrapper
notification-orchestrator.ts - Fan-out: push, email, SMS
db.ts                 - Drizzle + pg Pool singleton

```

Database (`lib/db/src/`)

```

schema/schema.ts      - All Drizzle table definitions (2,377 lines)
index.ts              - Re-exports all tables and types

```

Frontend (`artifacts/myhomebase/src/`)

```

pages/contractor-dashboard.tsx - 2011 lines - Main contractor hub
pages/contractor-crm.tsx       - 2627 lines - Full CRM UI
pages/contractor-onboarding.tsx - 572 lines - 4-step registration wizard
components/proposals.tsx       - 873 lines - Proposal CRUD + e-signature
components/stripe-connect-onboarding.tsx - 293 lines - Stripe Connect setup
components/contractor-feature-gate.tsx - 338 lines - Pro-tier paywall gate

```

5. Frontend Architecture

contractor-dashboard.tsx — Exported identifiers

```

TYPESCRIPT
export default function ContractorDashboard() {

```

contractor-crm.tsx — Interfaces (type contracts)

```

TYPESCRIPT
interface CrmLead {
interface CrmIntegration {
interface CrmClient {
interface CrmJob {
interface QuoteLineItem {
interface CrmQuote {
interface CrmInvoice {
interface DashboardStats {

```

contractor-onboarding.tsx — Form steps and services list (head)


```

TYPESCRIPT
import { useState } from "react";
import { useLocation } from "wouter";
import { useMutation } from "@tanstack/react-query";
import { useToast } from "@hooks/use-toast";
import { Check, Briefcase, Wrench, Shield, Rocket, ChevronRight, ChevronLeft, Plus, X } from "lucide-react";
import "./home.css";

const C = {
  primary: '#1560A2',
  deep: '#0C3460',
  tint: '#EAF4FD',
  eyebrow: '#AFD6F9',
  border: 'rgba(21,96,162,0.12)',
};

const SERVICES = [
  "Appliance Installation", "Appliance Repair & Maintenance", "Basement Remodeling",
  "Bathroom Remodeling", "Cabinet Installation", "Carpet Cleaning", "Carpet Installation",
  "Chimney & Fireplace Services", "Concrete & Masonry", "Custom Carpentry",
  "Deck Construction", "Drainage Solutions", "Drywall & Spackling Repair",
  "Electrical Services", "Epoxy Flooring", "Exterior Painting", "Fence Installation",
  "Fire & Water Damage Restoration", "Furniture Assembly", "Garage Door Services",
  "General Contracting", "Gutter Cleaning and Repair", "Gutter Installation",
  "Handyman Services", "Hardwood Flooring", "Home Automation & Tech Services",
  "Home Inspection", "House Cleaning", "HVAC Services", "Interior Painting",
  "Irrigation Systems", "Junk Removal", "Kitchen Remodeling", "Laminate & Vinyl Flooring",
  "Landscape Design", "Lawn & Landscaping", "Local Moving", "Locksmiths",
  "Mold Remediation", "Pest Control", "Plumbing Services", "Pool Installation",
  "Pool Maintenance", "Pressure Washing", "Roofing Services",
  "Security System Installation", "Septic Services", "Siding Installation",
  "Snow Removal", "Tile Installation", "Tree Service & Trimming",
  "Trim & Finish Carpentry", "Window Cleaning", "Windows & Door Installation",
];

const EXPERIENCE_OPTIONS = ["1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12", "13", "14", "15", "16", "17", "18", "19", "20", "20+"];

const inpStyle: React.CSSProperties = {
  width: '100%', background: '#F3F5F7', border: `1.5px solid ${C.border}`,
  borderRadius: 10, padding: '12px 14px', fontSize: 13, fontWeight: 500,
  color: '#1a1a1a', boxSizing: 'border-box', fontFamily: 'inherit',
  outline: 'none', appearance: 'none' as const,
};

const labelStyle: React.CSSProperties = {
  fontSize: 11, fontWeight: 700, color: C.primary,
  letterSpacing: '0.03em', marginBottom: 6, display: 'block',
};

const STEPS = [
  { icon: Briefcase, label: "Basics" },
  { icon: Wrench, label: "Services" },
  { icon: Shield, label: "Credentials" },
  { icon: Rocket, label: "Done!" },
];

type FormState = {

```

contractor-feature-gate.tsx — Feature keys (from live source)


```

TYPESCRIPT
const featureLabels: Record<string, { label: string; icon: React.ReactNode; description: string }> = {
  crm: { label: 'CRM Features', icon: <Users className="h-5 w-5" />, description: 'Full customer relationship management' },
  clients: { label: 'Client Management', icon: <Users className="h-5 w-5" />, description: 'Manage your customer database' },
  jobs: { label: 'Job Scheduling', icon: <Calendar className="h-5 w-5" />, description: 'Schedule and track jobs' },
  notes: { label: 'Quotes & Estimates', icon: <FileText className="h-5 w-5" />, description: 'Create professional quotes' },
  invoicing: { label: 'Invoicing', icon: <FileText className="h-5 w-5" />, description: 'Send invoices and track payments' },
  payments: { label: 'Payment Processing', icon: <CreditCard className="h-5 w-5" />, description: 'Accept payments via Stripe' },
  team: { label: 'Team Management', icon: <Users className="h-5 w-5" />, description: 'Manage your team members' },
  imports: { label: 'Data Import', icon: <Download className="h-5 w-5" />, description: 'Import from other CRMs' },
  analytics: { label: 'Business Analytics', icon: <BarChart3 className="h-5 w-5" />, description: 'Detailed business insights' },
};

```

stripe-connect-onboarding.tsx — Status interface (from live source)

```

TYPESCRIPT
interface StripeConnectStatus {
  hasAccount: boolean;
  accountId?: string;
  onboardingComplete: boolean;
  chargesEnabled: boolean;
  payoutsEnabled: boolean;
}

```

6. API Reference — Public /api/contractors

Extracted from routes.ts — 14 routes

Method	Path	Auth Middleware
GET	/api/contractors	none
GET	/api/contractors/search	none
GET	/api/contractors/previously-used	isAuthenticated
GET	/api/contractors/:id	none
POST	/api/contractors/boost	isAuthenticated
GET	/api/contractors/boost/check	isAuthenticated
DELETE	/api/contractors/boost/:boostId	isAuthenticated
GET	/api/contractors/:contractorId/contacted-homeowners	isAuthenticated
GET	/api/contractors/:id/reviews	none
GET	/api/contractors/:id/rating	none
GET	/api/contractors/:id/can-review	isAuthenticated
GET	/api/contractors/:id/eligible-service-records	isAuthenticated
POST	/api/contractors/:id/reviews	isAuthenticated
POST	/api/contractors/:id/review-request	isAuthenticated

7. API Reference — /api/contractor Management

Extracted from routes.ts — 35 routes

Method	Path	Auth Middleware
POST	/api/contractor/stripe-connect/create	isAuthenticated, requireRole
POST	/api/contractor/stripe-connect/onboarding-link	isAuthenticated, requireRole
GET	/api/contractor/stripe-connect/status	isAuthenticated, requireRole

POST	/api/contractor/stripe-connect/dashboard-link	isAuthenticated, requireRole
POST	/api/contractor/upload-logo	isAuthenticated
GET	/api/contractor/notifications/preferences	isAuthenticated
PATCH	/api/contractor/notifications/preferences	isAuthenticated
GET	/api/contractor/subscription	isAuthenticated
GET	/api/contractor/my-home	isAuthenticated
POST	/api/contractor/my-home	isAuthenticated
PATCH	/api/contractor/my-home/:houseId	isAuthenticated
DELETE	/api/contractor/my-home/:houseId	isAuthenticated
GET	/api/contractor/my-home/tasks	isAuthenticated
GET	/api/contractor/profile	isAuthenticated, requireContractorSubscription
PUT	/api/contractor/profile	requireContractorSubscription
GET	/api/contractor/licenses	isAuthenticated
POST	/api/contractor/licenses	isAuthenticated
PUT	/api/contractor/licenses/:id	isAuthenticated
DELETE	/api/contractor/licenses/:id	isAuthenticated
GET	/api/contractor/validate-token	none
POST	/api/contractor/accept-invite	none
POST	/api/contractor/invite-tech	isAuthenticated, requireCompanyRole, requireNotSuspended
POST	/api/contractor/team/:userId/resend-invite	isAuthenticated, requireCompanyRole, requireNotSuspended
GET	/api/contractor/team	isAuthenticated, requireCompanyRole, requireNotSuspended
PATCH	/api/contractor/team/:userId/suspend	isAuthenticated, requireCompanyRole, requireNotSuspended, requireSameCompany
PATCH	/api/contractor/team/:userId/reactivate	isAuthenticated, requireCompanyRole, requireNotSuspended, requireSameCompany
DELETE	/api/contractor/team/:userId/invite	isAuthenticated, requireCompanyRole, requireNotSuspended, requireSameCompany
PATCH	/api/contractor/team/:userId	isAuthenticated, requireCompanyRole, requireNotSuspended, requireSameCompany
DELETE	/api/contractor/team/:userId	isAuthenticated, requireCompanyRole, requireNotSuspended, requireSameCompany
GET	/api/contractor/team/audit-log	isAuthenticated, requireCompanyRole, requireNotSuspended
GET	/api/contractor/team/:userId/audit-log	isAuthenticated, requireCompanyRole, requireNotSuspended
GET	/api/contractor/company-homeowners	isAuthenticated, requireCompanyRole, requireNotSuspended
POST	/api/contractor/invoices/upload	isAuthenticated, requireCompanyRole, requireNotSuspended
GET	/api/contractor/invoices	isAuthenticated, requireCompanyRole, requireNotSuspended
GET	/api/contractor/invoices/:id	isAuthenticated, requireNotSuspended

8. API Reference — CRM /api/crm/

Extracted from routes.ts — 41 routes

Method	Path	Auth Middleware
POST	/api/crm/invoices/:invoiceId/payment-link	isAuthenticated, requireRole
GET	/api/crm/leads	isAuthenticated, requireContractorSubscription
POST	/api/crm/leads	isAuthenticated, requireContractorSubscription
GET	/api/crm/leads/:id	isAuthenticated, requireContractorSubscription
PATCH	/api/crm/leads/:id	isAuthenticated
DELETE	/api/crm/leads/:id	isAuthenticated
POST	/api/crm/leads/:leadId/notes	isAuthenticated

PATCH	/api/crm/notes/:id	isAuthenticated
DELETE	/api/crm/notes/:id	isAuthenticated
GET	/api/crm/integrations	isAuthenticated
POST	/api/crm/integrations	isAuthenticated
DELETE	/api/crm/integrations/:id	isAuthenticated
POST	/api/crm/webhooks/:integrationId	none
GET	/api/crm/webhooks/:integrationId/logs	isAuthenticated
GET	/api/crm/clients	isAuthenticated, requireContractorSubscription
POST	/api/crm/clients	isAuthenticated, requireContractorSubscription
GET	/api/crm/clients/:id	isAuthenticated
PATCH	/api/crm/clients/:id	isAuthenticated
DELETE	/api/crm/clients/:id	isAuthenticated
GET	/api/crm/jobs	isAuthenticated
POST	/api/crm/jobs	isAuthenticated
GET	/api/crm/jobs/:id	isAuthenticated
PATCH	/api/crm/jobs/:id	isAuthenticated
DELETE	/api/crm/jobs/:id	isAuthenticated
POST	/api/crm/jobs/:id/notify	isAuthenticated
GET	/api/crm/quotes	isAuthenticated
POST	/api/crm/quotes	isAuthenticated
GET	/api/crm/quotes/:id	isAuthenticated
PATCH	/api/crm/quotes/:id	isAuthenticated
POST	/api/crm/quotes/:id/send	isAuthenticated
DELETE	/api/crm/quotes/:id	isAuthenticated
GET	/api/crm/invoices	isAuthenticated
POST	/api/crm/invoices	isAuthenticated
GET	/api/crm/invoices/:id	isAuthenticated
PATCH	/api/crm/invoices/:id	isAuthenticated
POST	/api/crm/invoices/:id/send	isAuthenticated
POST	/api/crm/invoices/:id/payment	isAuthenticated
DELETE	/api/crm/invoices/:id	isAuthenticated
GET	/api/crm/dashboard	isAuthenticated, requireContractorSubscription
POST	/api/crm/import	isAuthenticated
GET	/api/crm/import/template	isAuthenticated

9. API Reference — /api/proposals

Extracted from routes.ts — 7 routes

Method	Path	Auth Middleware
GET	/api/proposals	isAuthenticated
GET	/api/proposals/:id	isAuthenticated
POST	/api/proposals	isAuthenticated
PATCH	/api/proposals/:id	isAuthenticated
POST	/api/proposals/:id/sign	isAuthenticated

POST	/api/proposals/:id/contract	isAuthenticated
DELETE	/api/proposals/:id	isAuthenticated

Proposal Status Flow

```

draft !' sent !' accepted !' (contract uploaded) !' (contract signed)
      !' rejected
      !' expired

```

10. Database Schema

Extracted from lib/db/src/schema/schema.ts — 10 tables

`companies` (`companies`)

Column	Type	Notes
id	uuid/varchar	auto
name	text	not null
bio	text	not null
experience	int	not null, default: 0
location	text	not null
ownerId	varchar	not null
rating	decimal	not null, default: "0"
reviewCount	int	not null, default: 0
services	text[]	not null
phone	text	not null
email	text	not null
address	text	
city	text	
state	text	
postalCode	text	
latitude	decimal	
longitude	decimal	
serviceRadius	int	not null, default: 25
hasEmergencyServices	bool	not null, default: false
businessLogo	text	
projectPhotos	text[]	default: sqlARRAY[]::text[]
website	text	
facebook	text	
instagram	text	
linkedin	text	
googleBusinessUrl	text	
countryId	varchar	FK !' countries
regionId	varchar	FK !' regions
licenseNumber	text	not null
licenseMunicipality	text	not null

isLicensed	bool	not null, default: true
licenses	text	
insuranceInfo	text	
referralCode	varchar	unique
stripeConnectAccountId	varchar	
stripeOnboardingComplete	bool	default: false
stripeChargesEnabled	bool	default: false
stripePayoutsEnabled	bool	default: false
stripeDefaultCurrency	varchar	default: "usd"
subscriptionTier	text	default: "individual"
maxTechSeats	int	default: 3
createdAt	timestamp	auto
updatedAt	timestamp	auto

`company\_invite\_codes` (`companyInviteCodes`)

Column	Type	Notes
id	uuid/varchar	auto
companyId	varchar	FK !' companies, not null
code	varchar	not null, unique
createdBy	varchar	not null
isActive	bool	not null, default: true
usedBy	varchar	
usedAt	timestamp	
expiresAt	timestamp	
createdAt	timestamp	auto

`contractors` (`contractors`)

Column	Type	Notes
id	uuid/varchar	auto
userId	varchar	not null
companyId	varchar	FK !' companies, not null
name	text	not null
company	text	not null
bio	text	not null
location	text	not null
distance	decimal	
rating	decimal	not null
reviewCount	int	not null, default: 0
experience	int	not null
services	text[]	not null
phone	text	not null
email	text	not null
address	text	
city	text	
state	text	

licenseNumber	text	not null
licenseMunicipality	text	not null
isLicensed	bool	not null, default: true
serviceRadius	int	not null, default: 25
hasEmergencyServices	bool	not null, default: false
profileImage	text	
businessLogo	text	
projectPhotos	text[]	default: sqlARRAY[]::text[]
isVerified	bool	not null, default: false
insuranceCarrier	text	
insurancePolicyNumber	text	
insuranceExpiryDate	text	
insuranceCoverageAmount	text	
website	text	
facebook	text	
instagram	text	
linkedin	text	
googleBusinessUrl	text	
countryId	varchar	FK ! ' countries
regionId	varchar	FK ! ' regions
licenses	text	
insuranceInfo	text	
postalCode	text	
createdAt	timestamp	auto

`contractor\_reviews` (`contractorReviews`)

Column	Type	Notes
id	uuid/varchar	auto
contractorId	text	not null
companyId	varchar	FK ! ' companies
homeownerId	text	not null
rating	int	not null
comment	text	
serviceDate	timestamp	
serviceType	text	
wouldRecommend	bool	not null, default: true
deviceFingerprint	text	
ipAddress	varchar	
isVerifiedService	bool	not null, default: false
serviceRecordId	varchar	
reviewPhotoUrl	text	
contractorResponse	text	
contractorRespondedAt	timestamp	
createdAt	timestamp	auto
updatedAt	timestamp	auto

`proposals` (`proposals`)

Column	Type	Notes
id	uuid/varchar	auto
contractorId	text	not null
companyId	varchar	FK !' companies
createdBy	varchar	
homeownerId	text	
title	text	not null
description	text	not null
serviceType	text	not null
estimatedCost	decimal	not null
estimatedDuration	text	not null
scope	text	not null
materials	text[]	not null, default: sql '{}':text[]
warrantyPeriod	text	
validUntil	text	not null
status	text	not null, default: "draft"
customerNotes	text	
internalNotes	text	
attachments	text[]	default: sql '{}':text[]
contractFilePath	text	
contractSignedAt	timestamp	
customerSignature	text	
contractorSignature	text	
signatureIpAddress	text	
createdAt	timestamp	auto
updatedAt	timestamp	auto

`contractor\_boosts` (`contractorBoosts`)

Column	Type	Notes
id	uuid/varchar	auto
contractorId	text	not null
serviceCategory	text	not null
businessAddress	text	not null
businessLatitude	decimal	not null
businessLongitude	decimal	not null
boostRadius	int	not null, default: 10
startDate	timestamp	not null
endDate	timestamp	not null
amount	decimal	not null
stripePaymentIntentId	text	
status	text	not null, default: "active"
isActive	bool	not null, default: true
createdAt	timestamp	auto

updatedAt	timestamp	auto
stripePaymentIntentIdUnique	text	

`crm\_clients` (`crmClients`)

Column	Type	Notes
id	uuid/varchar	auto
contractorUserId	varchar	FK !' users, not null
companyId	varchar	FK !' companies
firstName	text	not null
lastName	text	not null
email	text	
phone	text	
secondaryPhone	text	
address	text	
city	text	
state	text	
postalCode	text	
notes	text	
tags	text[]	default: sqlARRAY[]::text[]
preferredContactMethod	text	default: "phone"
isActive	bool	not null, default: true
totalJobsCompleted	int	not null, default: 0
totalRevenue	decimal	default: "0.00"
lastServiceDate	timestamp	
createdAt	timestamp	auto
updatedAt	timestamp	auto

`crm\_jobs` (`crmJobs`)

Column	Type	Notes
id	uuid/varchar	auto
contractorUserId	varchar	FK !' users, not null
companyId	varchar	FK !' companies
clientId	varchar	FK !' crmClients, not null
quoteId	varchar	
title	text	not null
description	text	
serviceType	text	not null
status	text	not null, default: "scheduled"
priority	text	not null, default: "normal"
scheduledDate	timestamp	not null
scheduledEndDate	timestamp	
actualStartTime	timestamp	
actualEndTime	timestamp	
estimatedDuration	int	
actualDuration	int	

address	text	
city	text	
state	text	
postalCode	text	
laborCost	decimal	
materialsCost	decimal	
totalCost	decimal	
notes	text	
internalNotes	text	
completionNotes	text	
createdAt	timestamp	auto
updatedAt	timestamp	auto

`crm\_quotes` (`crmQuotes`)

Column	Type	Notes
id	uuid/varchar	auto
contractorUserId	varchar	FK !' users, not null
companyId	varchar	FK !' companies
clientId	varchar	FK !' crmClients, not null
quoteNumber	varchar	not null
title	text	not null
description	text	
serviceType	text	not null
status	text	not null, default: "draft"
lineItems	jsonb	not null, default: sql[]::jsonb
subtotal	decimal	not null
taxRate	decimal	default: "0.00"
taxAmount	decimal	default: "0.00"
discount	decimal	default: "0.00"
total	decimal	not null
validUntil	timestamp	
sentAt	timestamp	
viewedAt	timestamp	
acceptedAt	timestamp	
declinedAt	timestamp	
notes	text	
termsAndConditions	text	
createdAt	timestamp	auto
updatedAt	timestamp	auto

`crm\_invoices` (`crmInvoices`)

Column	Type	Notes
id	uuid/varchar	auto
contractorUserId	varchar	FK !' users, not null
companyId	varchar	FK !' companies

clientId	varchar	FK !' crmClients, not null
jobId	varchar	FK !' crmJobs
quoteId	varchar	FK !' crmQuotes
invoiceNumber	varchar	not null
title	text	not null
description	text	
status	text	not null, default: "draft"
lineItems	jsonb	not null, default: sql[]::jsonb
subtotal	decimal	not null
taxRate	decimal	default: "0.00"
taxAmount	decimal	default: "0.00"
discount	decimal	default: "0.00"
total	decimal	not null
amountPaid	decimal	default: "0.00"
amountDue	decimal	not null
dueDate	timestamp	
sentAt	timestamp	
viewedAt	timestamp	
paidAt	timestamp	
paymentMethod	text	
paymentNotes	text	
notes	text	
termsAndConditions	text	
homeownerId	varchar	FK !' users
houseId	varchar	FK !' houses
createdAt	timestamp	auto
updatedAt	timestamp	auto

11. Annotated Code Walkthroughs

11.1 Stripe Connect Initialization (from routes.ts)

```

TYPESCRIPT
const stripe = process.env.STRIPE_SECRET_KEY
  ? new Stripe(process.env.STRIPE_SECRET_KEY, { apiVersion: "2025-08-27.basil" })
  : null;

```

Stripe instance is null when STRIPE_SECRET_KEY is absent. All Stripe routes must null-check before use.

11.2 Proposal Component — Signature Flow


```

TYPESCRIPT
const signatureMutation = useMutation({
  mutationFn: ({ proposalId, signatureData }: {
    proposalId: string;
    signatureData: { signature: string; signerName: string; signedAt: string; ipAddress?: string }
  }) => apiRequest(`/api/proposals/${proposalId}/sign`, "POST", signatureData),
  onSuccess: () => {
    queryClient.invalidateQueries({ queryKey: ["/api/proposals"] });
    setShowSignature(false);
    setSigningProposal(null);
    toast({
      title: "Success",
      description: "Contract signed successfully",
    });
  },
  onError: () => {
    toast({
      title: "Error",
      description: "Failed to sign contract",
      variant: "destructive",
    });
  },
});

```

11.3 Stripe Connect Component — State Machine

Three display states driven by StripeConnectStatus:

```

!hasAccount           !' show "Connect with Stripe" CTA
hasAccount && !charges !' show "Complete Setup" (incomplete onboarding)
hasAccount && charges  !' show Connected + Stripe Express dashboard link

```

```

TYPESCRIPT
if (!status?.hasAccount) {
  return (
    <Card className="border-2 border-dashed border-blue-200">
      <CardHeader className="text-center">
        <div className="mx-auto mb-4 p-4 rounded-full bg-blue-100">

```

11.4 Feature Gate — hasCrmAccess Logic

```

TYPESCRIPT
export function ContractorFeatureGate({ children, feature, fallback }: ContractorFeatureGateProps) {
  const { hasCrmAccess, isLoading } = useContractorSubscription();

  if (isLoading) {
    return <div className="animate-pulse bg-muted h-32 rounded-lg" />;
  }

  if (hasCrmAccess) {
    return <{children}</>;
  }

  if (fallback) {
    return <{fallback}</>;
  }

  return <ContractorUpgradePrompt feature={feature} />;
}

```

12. Integration Points

Integration	Used by	Config env var
Stripe Connect	/api/contractor/stripe-connect/*	STRIPE_SECRET_KEY

Stripe Payment Links	/api/crm/invoices/:id/payment-link	STRIPE_SECRET_KEY
Stripe Subscriptions	Webhooks at /api/stripe/webhook	STRIPE_WEBHOOK_SECRET
SendGrid	Quote/invoice send, invite emails	SENDGRID_API_KEY
Twilio	Job notifications, quote/invoice SMS	TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN
OpenAI	Invoice PDF analysis (GPT-4o Vision)	OPENAI_API_KEY
Replit Object Storage	Logos, photos, proposal attachments	managed

13. Key Patterns & Gotchas

1. Company vs. Contractor Duality

Two tables exist: contractors (individual) and companies (team entity). Profile PUT auto-creates a company when companyId is null. Homeowner-facing queries prefer company data (logo, reviews) over individual contractor data.

2. Proposal Creation Restricted to Conversations

The Create Proposal <DialogTrigger> in proposals.tsx is inside {false && ...} — intentionally hidden. Proposal creation only surfaces in the Messages tab, scoped to active conversations, preventing cold spam.

3. Tech Role Reduced Capabilities

Techs (companyRole === 'tech') see <TechDashboard> instead of the full dashboard. Techs can upload invoices and view their own history only.

4. CRM Subscription Gap

GET /api/crm/clients requires requireContractorSubscription, but individual GET/PATCH/DELETE /api/crm/clients/:id only have isAuthenticated. Expired contractors can still read/modify individual records they already created.

5. No Pagination on List Endpoints

All list endpoints return full unbounded sets. Cursor-based pagination is needed before contractor/CRM data grows large.

6. Emergency Reset Endpoint

POST /api/admin/emergency-reset still exists in routes.ts (search for emergency-reset). EMERGENCY_RESET_SECRET has been removed from production env (returns 404), but the dead code should be removed in a future cleanup pass.

7. Stripe API Version Pinned

apiVersion: '2025-08-27.basil' is pinned in routes.ts. Any Stripe SDK upgrade must be checked against this version's breaking changes.

8. Rate Limits

authLimiter: 100/15min. uploadLimiter: 10/hour. aiChatLimiter: 20/hour. All applied per-IP.